

AMAN LONARE

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SUMMARY

AI Engineer with 5+ years of experience building production-grade ML & LLM systems. Specializes in end-to-end AI platforms, including scalable RAG systems, agentic workflows, & MLOps for reliable, high-performance deployment.

TECHNICAL SKILLS

Programming & Development: Python, Go, SQL, Bash, Dart, React, FastAPI, Flask, Cursor, Claude Code
AI and ML: TensorFlow, PyTorch, Hugging Face, ETL Pipeline, A/B Testing, RLHF, DSPy, PydanticAI
LLMs: LangGraph, RAG, Context Engineering, Agentic AI, MCP, Vector/Graph DB, DPO/PEFT, OpenAI
MLOps: AWS SageMaker & Bedrock, MLflow, Airflow, CI/CD, Model Monitoring, LLMops, Celery, LangSmith
Cloud & Tools: Docker, Kubernetes, Terraform, AWS EMR, Git, Vercel, Apache Spark, Flutter
Communication: English (Fluent), Hindi (Fluent), Japanese (N5, *N4 candidate (Expected [06/2026])*)

PROFESSIONAL EXPERIENCE (5+ YEARS)

AI Engineer | Archetype Studio, JPN | Full Time

Aug 2023 – Present

- **Project: Multi-Agent Orchestration for End-to-End Engineering (*in progress*)**
 - Architected a production-grade **multi-agent system** using **LangGraph** and **AWS Bedrock** with hierarchical Supervisor-Worker orchestration for planning, coding, ops, researcher, and growth workflows
 - Engineered **layout-aware chunking pipeline** using **Docling** and **Tree-sitter** for 5K+ PDFs & source code
 - Designed **hybrid RAG pipeline** using **Pinecone** & **Neo4j**, combining semantic retrieval, metadata filtering, dependency-aware context expansion, & **Qwen-based cross-encoder reranking** to reduce hallucinations
 - Built **self-corrective RAG**, reducing hallucinations with validation and refinement, achieving strong **RAG evaluation performance** using **RAGAS** (faithfulness: 92%, context recall: 92%) across 5K+ document dataset
 - Established robust **evaluation pipelines** & integrated **Langfuse** for observability, & monitoring of agents
- **Project: AI Copilot for Internal FinTech Platform (Client: Omise Japan)**
 - Architected & implemented core developer pipelines enabling end-to-end AI workflows, including **Advanced RAG** ingestion, **API & schema** tooling, & **execution runners**, leveraging **deep agents** for agentic workflows
 - Integrated AI coding assistants into the development workflow to accelerate scaffolding and deployment
 - Engineered modular prompt templates, **guardrails**, adapters, & test oracles for reliability & reproducibility
 - Implemented observability and evaluation dashboards to track latency, errors, token usage, and success rates
- **Project: Production-Scale AI System for Mobile Model Optimization & Deployment**
 - Deployed production ready model & AI platform, scaling to **1M+ daily** predictions for **500K+ users**
 - Introduced **MLOps culture** by setting up CI/CD pipelines with **Amazon SageMaker** tools, reducing integration time for new models from days to hours & increasing deployment frequency from bi-annual to monthly
 - Owned end-to-end system design from data ingestion to scalable model deployment & monitoring in production
 - Performed feature engineering and hyperparameter tuning (SageMaker HPO) to optimize model performance
 - Achieved **40% faster training** speeds by implementing **distributed training** using **SageMaker DDP**
 - Cut incident response time by 50% through custom monitoring system built with SageMaker **Model Monitor**
 - Improved model accuracy by 23% & reduced corrections by 25% through semi-automated continuous retraining
- **Project: AI-Driven Edge Computing System for Multi-Modal Transportation Classification**
 - Designed & developed an **edge-based transportation mode detection system** for iOS platform serving **500K+ users** with **1M+ daily** predictions, achieving **99.3% accuracy** in classifying 15 transportation modes
 - Optimized model footprint by 90% through hierarchical ML architecture while maintaining inference latency
 - Built cross-platform ML pipeline with m2cgen to convert Python models to native Dart, preserving performance
 - Reduced AWS costs by 70% by migrating ML inference to edge devices via **Flutter implementation**
 - Reduced operational costs 60% by implementing local feedback caching to eliminate OpenStreetMap API calls

AI Engineer | Hypothesis AI, IND | Consulting

May 2025 – July 2025

- **Project: Multilingual AI Voice Agent for Abandoned Checkout Recovery in E-commerce**
 - Designed scalable **multilingual AI voice agent** using OpenAI & RAG, achieving latency in milliseconds
 - Engineered multilingual (6 languages), context-aware prompts, boosting abandoned checkout recovery by 90%

- Developed event-driven integration framework with FastAPI, connecting Shopify with third party services and analytics platform (mixpanel), enabling seamless omni-channel customer engagement

AI Engineer (Founding Team) | MindYogi.me, IND | Consulting

Dec 2025 – Jan 2026

- Project: Advanced RAG-Enabled Chatbot and Voice Interface for Personalized Wellness Content

- Developed a scalable serverless architecture for a mental health platform (MindYogi), implementing AI-driven content generation workflows and **text-to-speech synthesis** powered by state-of-the-art models
- Designed and implemented an **advanced agentic RAG chatbot** using LangGraph, incorporating **self-reflective, corrective, and adaptive** retrieval strategies to improve grounding and minimize hallucinations
- Optimized the retrieval pipeline on Pinecone using recursive, context-enriched hybrid chunking, with reranking, caching, and query rewriting/expansion nodes, with continuous grounding and relevance evaluation via RAGAS
- Optimized a 10+ node LangGraph pipeline with advanced inference techniques, reducing TTFT to <5 seconds
- Built a low-latency **voice agent** using Fennec ASR, Baseten inference, and Inworld TTS, achieving <2s end-of-speech to first audio frame latency at around \$0.50 per hour of voice output
- Deployed real-time chatbot and voice inference as modular REST API services supporting multiple consumers, with low latency, monitoring, and production reliability

Research Software Engineer | Hitachi R&D, Tokyo, JPN | Full Time

Jan 2021 – July 2023

- Project: Distributed Data Management Framework for Microservice Architecture

- Developed tool for assisting the implementation of **CQRS & Event Sourcing** design patterns
- Reduced development time by 15% using **Domain Driven Design** software development approach
- Improved scalability & availability of the developed application using **Kafka & EventStoreDB** tools

***Patent:** System & method to assist modelling CQRS & ES based application [Submitted]*

Research Assistant | IIT Bombay, Mumbai, IND | Full Time

Sept 2020 – Nov 2020

- Department: Centre for Technology Alternatives for Rural Areas (CTARA)

- Developed a **web portal** for farmers to help them in assessing their crop in real-time using remote sensing
- Estimated and predicted the crop acreage by processing the spatiotemporal satellite images of the crop

PROJECTS

- AI-Powered Agentic Document Processing System for Automated Form Completion

- Developed an agentic, event-driven AI workflow with **LlamaIndex** for automated form filling tasks
- Integrated **Groq LLM** for dynamic extraction from both structured and unstructured documents
- Used **Hugging Face** embeddings for vector store indexing and efficient contextual document retrieval
- Implemented human-in-the-loop feedback system to iteratively refine agent accuracy and user alignment

- Multimodal AI System with Local LLMs for Document, Image, & Voice Processing

- Architected multimodal AI chatbot integrating **Ollama LLMs** and **ChromaDB** for local document processing
- Implemented semantic document search with **ChromaDB** vector storage for context-aware responses
- Enabled real-time audio transcription using **Whisper** models and designed intuitive UI with **Streamlit**
- Developed robust state management with thread-safe **SQLite** pooling, ensuring secure local deployment

- FarmGPT: MCP-Driven RAG & LLM Agricultural Advisor

- Architected modular decision support system using **Model Context Protocol (MCP)** for agriculture
- Integrated **RAG** servers and **ChromaDB** vector database for semantic search and document retrieval
- Engineered embeddings & prompt engineering for optimal LLM interaction with **PLLaMa** & agentic LLMs
- Implemented asynchronous SSE for real-time data streaming and API-first design for actionable farmer guidance

EDUCATION

Indian Institute of Technology Bombay | M.Tech in Technology & Development *Aug 2018 - July 2020*

- CGPA: 9.3/10 | Machine Learning in Remote Sensing | Advanced Statistics | Satellite Image Processing
- **Publications:** Lonare, A., Maheshwari, B., & Chinnsamy, P. (2022). Village level identification of sugarcane in Sangali, Maharashtra using open source data. Journal of Agrometeorology

Indian Institute of Technology Kanpur | B.Tech in Mechanical Engineering

July 2013 - Aug 2017